WORKED SOLUTION

STEP III 2017, Question 7

Pure • Tangents to an Ellipse

STEP III 2017, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q7

The key idea

The parameter t turns the condition "the tangent at T passes through (X, Y)" into a quadratic equation in t. Each root is a point of contact, so counting tangents from a point is the same as counting real roots, which is a discriminant question. In part (ii) the two contact parameters p and q are exactly the two roots of that quadratic, so their sum and product come straight from the coefficients.

Solution

The point lies on the ellipse

Substituting the coordinates of T,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{(1-t^2)^2 + 4t^2}{(1+t^2)^2} = \frac{1 + 2t^2 + t^4}{(1+t^2)^2} = 1,$$

so T lies on the ellipse for every real t.

Part (i)

Differentiating with respect to t,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{-4at}{(1+t^2)^2}, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{2b(1-t^2)}{(1+t^2)^2}.$$

Rather than divide these (awkward when $t = 0$), use the standard tangent to the ellipse at (x_0, y_0), namely $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$; its normal direction $\left(\frac{x_0}{a^2}, \frac{y_0}{b^2}\right)$ is perpendicular to the velocity above, since $\frac{1-t^2}{a} \cdot (-4at) + \frac{2t}{b} \cdot 2b(1-t^2) = 0$. At T the tangent L is therefore

$$\frac{x(1-t^2)}{a(1+t^2)} + \frac{2ty}{b(1+t^2)} = 1 \iff bx(1-t^2) + 2aty = ab(1+t^2).$$

Putting $(x, y) = (X, Y)$ and collecting powers of t:

$$bX - bXt^2 + 2aYt - ab - abt^2 = 0 \iff (a + X)bt^2 - 2aYt + b(a - X) = 0,$$

as required.

Two tangents. Read the other way round: the tangent at the point with parameter t passes through (X, Y) exactly when t satisfies this equation. As $X \neq -a$ and $b \neq 0$, the coefficient of t^2 is non-zero, so it is a genuine quadratic, with discriminant

$$4a^2Y^2 - 4b^2(a + X)(a - X) = 4(a^2Y^2 - (a^2 - X^2)b^2).$$

If $a^2Y^2 > (a^2 - X^2)b^2$ this is positive, so there are two distinct real roots $t_1 \neq t_2$. Different values of t give different points of the ellipse (the map $t \mapsto T$ is one-to-one; with $t = \tan\frac{\theta}{2}$ it is the familiar $(a\cos\theta, b\sin\theta)$ for $-\pi < \theta < \pi$). Different points of contact give different tangent lines, because a tangent to an ellipse meets it only at its point of contact. So there are two distinct tangents through (X, Y).

Ask yourself Could we have missed a tangent that the parameter t does not reach?

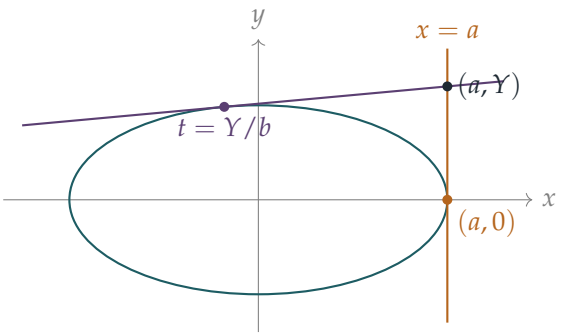
The only point of the ellipse not of the form $(*)$ is $(-a, 0)$ (the limit $t \to \infty$), whose tangent is $x = -a$. That line passes through (X, Y) only if $X = -a$, which is excluded. So the quadratic accounts for every tangent through (X, Y).

Interpretation. Dividing the condition by $a^2b^2 > 0$ gives

$$\frac{Y^2}{b^2} > 1 - \frac{X^2}{a^2} \iff \frac{X^2}{a^2} + \frac{Y^2}{b^2} > 1,$$

which says (X, Y) lies *outside* the ellipse. So from any point outside the ellipse we can draw two tangents to it, just as for a circle.

The case $X^2 = a^2$. Now the condition reads $a^2Y^2 > 0$, i.e. $Y \neq 0$: the point lies on one of the vertical lines $x = \pm a$, but not at the vertex $(\pm a, 0)$. One tangent is simply the vertical line $x = \pm a$ itself, which touches the ellipse at $(\pm a, 0)$. The other touches the ellipse somewhere else: for $X = a$ the quadratic becomes $2abt^2 - 2aYt = 0$, with roots $t = 0$ (the vertical tangent) and $t = Y/b \neq 0$; for $X = -a$ it becomes $-2aYt + 2ab = 0$, giving $t = b/Y$, and the second tangent is $x = -a$, the one at the missing point $(-a, 0)$. The sketch shows the case $X = a$, $Y > 0$.



So the result still holds when $X^2 = a^2$: there are two distinct tangents through (X, Y) whenever $Y \neq 0$, i.e. whenever the point is outside the ellipse.

Part (ii)

The tangent at P passes through (X, Y), so by part (i) p is a root of

$$(a + X)bt^2 - 2aYt + b(a - X) = 0,$$

and likewise q. Since $X \neq -a$ this is a quadratic, and $p \neq q$ (as $P \neq Q$), so p and q are its two roots. Hence

$$pq = \frac{b(a - X)}{b(a + X)} = \frac{a - X}{a + X} \quad \Longrightarrow \quad (a + X)pq = a - X,$$

as required, and

$$p + q = \frac{2aY}{b(a + X)}.$$

The final part

Setting $x = 0$ in the tangent $bx(1 - t^2) + 2aty = ab(1 + t^2)$ gives $2aty = ab(1 + t^2)$. The tangents do meet the y-axis, so $p, q \neq 0$ (at $t = 0$ the tangent is $x = a$), and

$$y_1 = \frac{b(1 + p^2)}{2p}, \qquad y_2 = \frac{b(1 + q^2)}{2q}.$$

Writing $\frac{1+p^2}{p} = p + \frac{1}{p}$,

$$y_1 + y_2 = \frac{b}{2}\left(p + q + \frac{1}{p} + \frac{1}{q}\right) = \frac{b}{2}(p + q)\left(1 + \frac{1}{pq}\right) = \frac{b(p + q)(1 + pq)}{2pq}.$$

From the results above, $1 + pq = \dfrac{2a}{a + X}$, so, using $pq = \dfrac{a - X}{a + X} \neq 0$,

$$y_1 + y_2 = \frac{b}{2} \cdot \frac{2aY}{b(a + X)} \cdot \frac{2a}{a + X} \cdot \frac{a + X}{a - X} = \frac{2a^2Y}{(a + X)(a - X)} = \frac{2a^2Y}{a^2 - X^2}.$$

Setting this equal to $2b$ and multiplying by $(a^2 - X^2)/(2a^2b)$ (non-zero, as $X^2 \neq a^2$):

$$\frac{Y}{b} = \frac{a^2 - X^2}{a^2} = 1 - \frac{X^2}{a^2} \iff \frac{X^2}{a^2} + \frac{Y}{b} = 1,$$

as required. So the intersection point lies on a parabola through $(\pm a, 0)$ and $(0, b)$.

Checks

- With $a = 2$, $b = 1$ and $(X, Y) = (4, 1)$ (outside, since $4 + 1 > 1$), the quadratic is $6t^2 - 4t - 2 = 0$, with roots $t = 1$ and $t = -\frac{1}{3}$: the tangents at $(0, 1)$ and $(1.6, -0.6)$. Both, $y = 1$ and $2x - 3y = 5$, do pass through $(4, 1)$. Here $pq = -\frac{1}{3} = \frac{2-4}{2+4}$ and $p + q = \frac{2}{3} = \frac{4}{6}$, as the formulas predict.

- At the centre $(0,0)$ the discriminant is $-4a^2b^2 < 0$: no tangents, as expected from inside the ellipse.
- Many random ellipses and pairs of points P, Q were checked by computer, intersecting the tangents directly; the formulas for pq, $p+q$ and y_1+y_2 all agreed, and when $y_1+y_2=2b$ the intersection satisfied $X^2/a^2+Y/b=1$.

Common slips

- Finding the gradient as $\frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}$ and forgetting that it fails at $t=0$, where the tangent is vertical.
- Treating $(a+X)bt^2-2aYt+b(a-X)=0$ as a quadratic when $X=-a$; this is exactly why the case $X^2=a^2$ needs separate treatment.
- Stopping at "the discriminant is positive" without saying why two roots give two *different* lines.

Beyond STEP

The rational parametrisation $\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$ of a conic is the start of algebraic geometry: it is why every conic with one rational point has infinitely many, and it underlies the classification of Pythagorean triples. The pair of tangents from an external point leads to the idea of the *polar* line and to projective geometry, where the awkward missing point at $t=\infty$ becomes an ordinary point. Siklos, *Advanced Problems in Mathematics*, has more questions of this type.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.